## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 2.1 · Completing the square

Consider the function $f$ defined on $\mathbb{R}$ by $f(x) = x^2 - 8x + 7$.

1. Complete the square to write $f(x)$ in the form $(x-4)^2 + k$, where $k$ is a real number to be determined.
2. Deduce a factorisation of $f(x)$ using a special product.
3. Draw the sign table of $f$, then solve the inequality $f(x) \leq 0$ in $\mathbb{R}$.

### Exercise 2.2 · Discussing without heavy calculation

We study the functions $f(x) = x^2 + 4$, $g(x) = (x+2)^2$ and $h(x) = -3(x-1)(x+4)$, all defined on $\mathbb{R}$.

1. For each function, determine any real roots by using the given form directly.
2. State the sign of each function on $\mathbb{R}$. State explicitly the values where it vanishes.
3. Explain why calculating a discriminant is necessary in none of the three cases.

### Exercise 2.3 · Choosing a strategy

We wish to solve the following three equations in $\mathbb{R}$:

$$x^2 - 9 = 0, \qquad x^2 - 9x + 20 = 0, \qquad 2x^2 - 3x - 1 = 0.$$

1. For each equation, choose a method: special product, finding two numbers with a known sum and product, or discriminant. Explain this choice.
2. Solve the three equations and give the solution sets in exact form.
3. Check the solutions of the first two equations by substitution. For the third, check the sum and the product of the roots obtained.

### Exercise 2.4 · Recovering a parabola

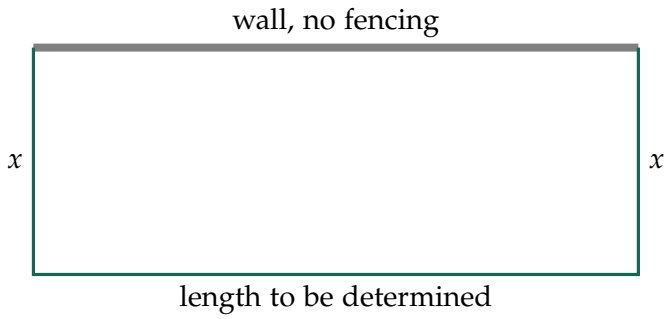
The curve of a polynomial function $f$ of degree 2 crosses the x-axis at the points with x-coordinates $-2$ and 3. It also passes through the point $A(1, 12)$.

1. Determine an expression for $f(x)$ from these three pieces of information. State the form chosen and why it is suitable.
2. Expand the expression found. Deduce the y-intercept of the curve.
3. Check that the function obtained satisfies the three initial pieces of information.

### Exercise 2.5 · A vegetable garden against a wall

A gardener has 20 m of fencing to enclose three sides of a rectangular vegetable garden. The fourth side runs along a wall and needs no fencing. Let $x$ be the length, in metres, of each of the two sides perpendicular to the wall. All the fencing is used.

1. Make a sketch and express the length of the side parallel to the wall in terms of $x$. State the values of $x$ for which the rectangle is not degenerate.
2. Express the area $A(x)$, in square metres. Determine its maximum value and the corresponding dimensions, using the method of your choice.
3. Check that these dimensions use exactly 20 m of fencing.




Sketch of the vegetable garden: only the three green lines use the 20 m of fencing. The proportions of the drawing do not fix the optimal dimensions.

### Exercise 2.6 · A parameter

For each real number $m$, consider the equation in the real unknown $x$:

$$x^2 - 2x + m = 0.$$

The number $m$ is a parameter: it stays fixed while the equation is solved.

1. Determine the values of $m$ for which the equation has two distinct real solutions.
2. Determine the values of $m$ for which it has a single real solution, then those for which it has none.
3. Give the solution in the case where it is unique. Explain why the three cases considered cover all values of $m$.

### Exercise 2.7 · Link with sequences

A sequence is defined by $u_n = n^2 - 5n + 4$ for every natural number $n$. We look for all the indices at which the term of the sequence is strictly negative.

1. First solve the inequality $x^2 - 5x + 4 < 0$ in $\mathbb{R}$.
2. Deduce all the natural numbers $n$ such that $u_n < 0$. Justify why a real interval is not the final answer here.

### Exercise 2.8 · Delayed review

A student studies the function $f$ defined on $\mathbb{R}$ by $f(x) = 2(x+3)^2 - 5$. She writes: "The minimum is $-5$ and it is reached at $x = 3$."

1. State what is correct and what is wrong in this sentence. Rewrite it correctly, justifying your answer.
2. Solve the inequality $f(x) \leq -3$ in $\mathbb{R}$, then check the two endpoints of the set obtained.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 43 p. 99, questions a and c.** First find an obvious root, then the other using the product of the roots.

**CQFD, exercise 49 p. 99, questions a and b.** Translate the sum and the product into an equation; do not assume in advance that the numbers exist.